I LEARN ACADEMY

وزارة الاتصالات وتكنولوجيا المعلومات

رواد مصر الرقمية

SUPPLY CHAIN

Optimization & Performance Overview.

R4-DATA ANALYST SPECIALIST ONL4_DAT1_S1

UNDER THE SUPERVISION OF

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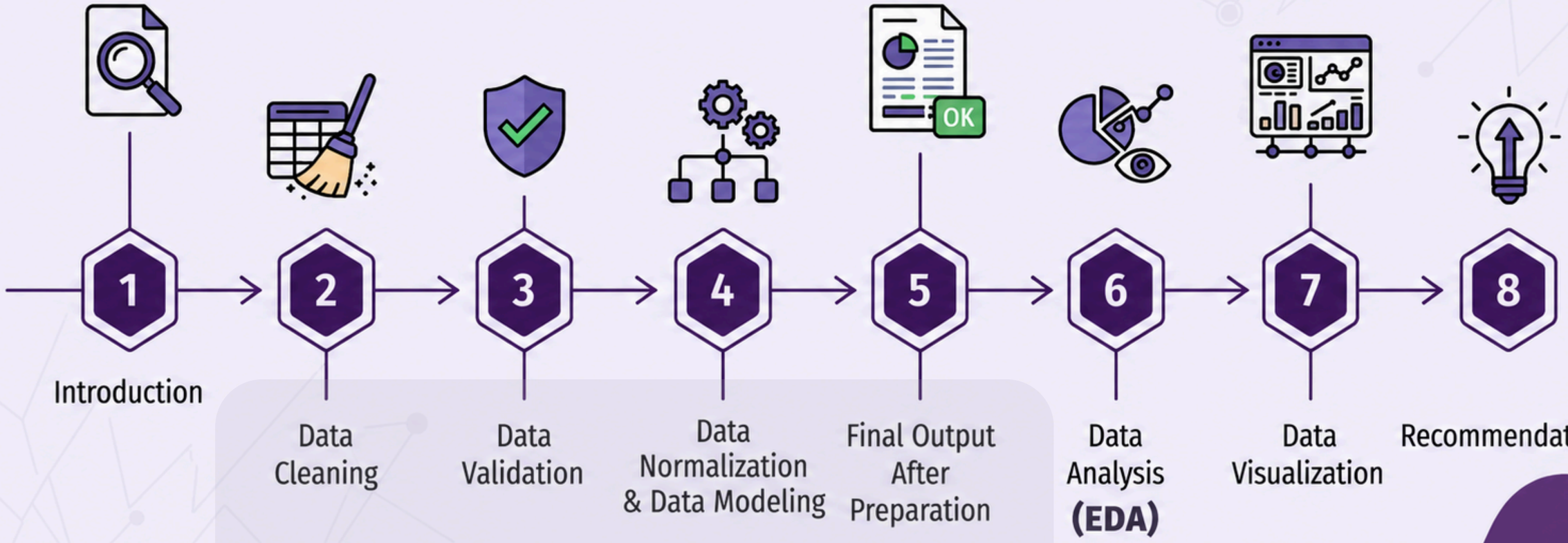
AGENDA



DATA PREPREPARATION &PREPROCESSING

INTRODUCTION

CLICK HERE

URL

SUPPLY CHAIN ILLUSTRATOR IS AN INTERACTIVE WEB APPLICATION DESIGNED TO SIMPLIFY SUPPLY CHAIN CONCEPTS THROUGH CLEAR VISUALIZATIONS AND PRACTICAL EXAMPLES. IT HELPS USERS UNDERSTAND THE FLOW OF PRODUCTS, INVENTORY, SUPPLIERS, AND LOGISTICS IN AN INTUITIVE WAY. THE PLATFORM PROVIDES AN ENGAGING LEARNING EXPERIENCE BY COMBINING VISUAL ILLUSTRATIONS WITH REAL-WORLD SUPPLY CHAIN SCENARIOS.

Problem Overview & Proposed Solution

KEY CHALLENGES IDENTIFIED

- Isolated reporting limits performance insight
- Frequent inventory imbalances
- Issues with supplier quality variation
- Operational bottlenecks causing shipping delays

COMPREHENSIVE ANALYTICS SOLUTION

- Platform consolidating data across all business domains
- Centralized, interactive dashboards for key stakeholders
- Real-time monitoring of critical KPIs and trends
- Data-driven decision-making across the supply chain

PROJECT UNIQUE VALUE PROPOSITION

- Integrated analytics beyond traditional single-area dashboards
- Connects Commercial, Supplier, Inventory, Shipping, & Customer metrics
- Enables root-cause analysis, not just symptom monitoring
- Drives proactive actions and measurable business impact

REVENUE	$182.81M	▼ Risk Indicator 2.1%
PROFIT	$85.63M	▼ Risk Indicator 3.2%
INVENTORY GAP	288K	▼ Risk Indicator 0.2%
RETURN RATE	7.38%	▼ Risk Indicator 2.1%

PROJECT OBJECTIVES

- Revenue & Profitability Analysis
- Customer Experience Evaluation
- Supplier Performance Assessment
- Inventory Performance Monitoring
- Logistics & Delivery Optimization
- Data-Driven Business Decision Support

Data Analytics Toolset

A Comprehensive Stack for Data Analysis

01

Microsoft Excel

Data manipulation, pivot tables, initial data exploration, and reporting.

02

SQL Server (Structured Query Language)

Querying relational databases (e.g., SQL Server) for data extraction and transformation.

03

Python

Advanced data cleaning, complex analysis, statistical modeling, and scripting.

04

Tableau

Creating interactive data visualizations, dashboards, and business intelligence.

05

Microsoft Power BI

Building dynamic reports, data modeling, and enterprise-wide BI solutions.

From Raw Data to Analysis-Ready Dataset

Data Preparation Workflow

From Raw Data to Analysis-Ready Dataset

01

Data Acquisition

Gather raw data from multiple business sources.

02

Data Cleaning

Remove duplicates, missing values, and formatting issues.

03

Data Validation

Verify data quality, consistency, and accuracy.

04

Data Modeling & Normalization

Organize the dataset into an optimized relational structure.

05

Analysis-Ready Dataset

Deliver a clean, validated, and structured dataset for analysis.

From Raw Data to Analysis-Ready Dataset

Data Acquisition

DATASET OVERVIEW

- This dataset represents the operations of an electronics **supply chain** business. It contains information about products, sales, inventory, suppliers, shipping, and customer satisfaction.

Structure:

- 1500 Rows
- 32 Columns
- 9 Feature Engineering (CALCULATED COLUMNS)

KEY BUSINESS METRICS DEFINITIONS

01

REVENUE

- Total income generated from sales.

02

PROFIT

- Revenue after deducting costs.

03

INVENTORY LEVEL

- Available stock quantity.

04

STOCKOUT RISK

- Probability of inventory shortage.

05

PROFIT MARGIN

- Percentage of profit earned relative to revenue. (CALCULATED COLUMN)

06

ON-TIME DELIVERY

- Orders delivered within the promised time.

07

CUSTOMER SATISFACTION SCORE

- Measures customer experience.

Data Cleaning

01

1. Remove Extra Spaces & Standardize Text

records standardized for text quality

```
update electronics_supplyChain
set
  Product_type = trim(Product_type),
  Category = trim(Category),
  Sales_Channel = trim(Sales_Channel),
  Payment_Method = trim(Payment_Method),
  Region = trim(Region),
  Supplier_name = trim(Supplier_name),
  Return_Reason = trim(Return_Reason),
  Customer_demographics = trim(Customer_demographics)
```

Description: Review data for leading, trailing, and multiple spaces. Standardize capitalization and dates.

02

2. Remove Duplicate Records

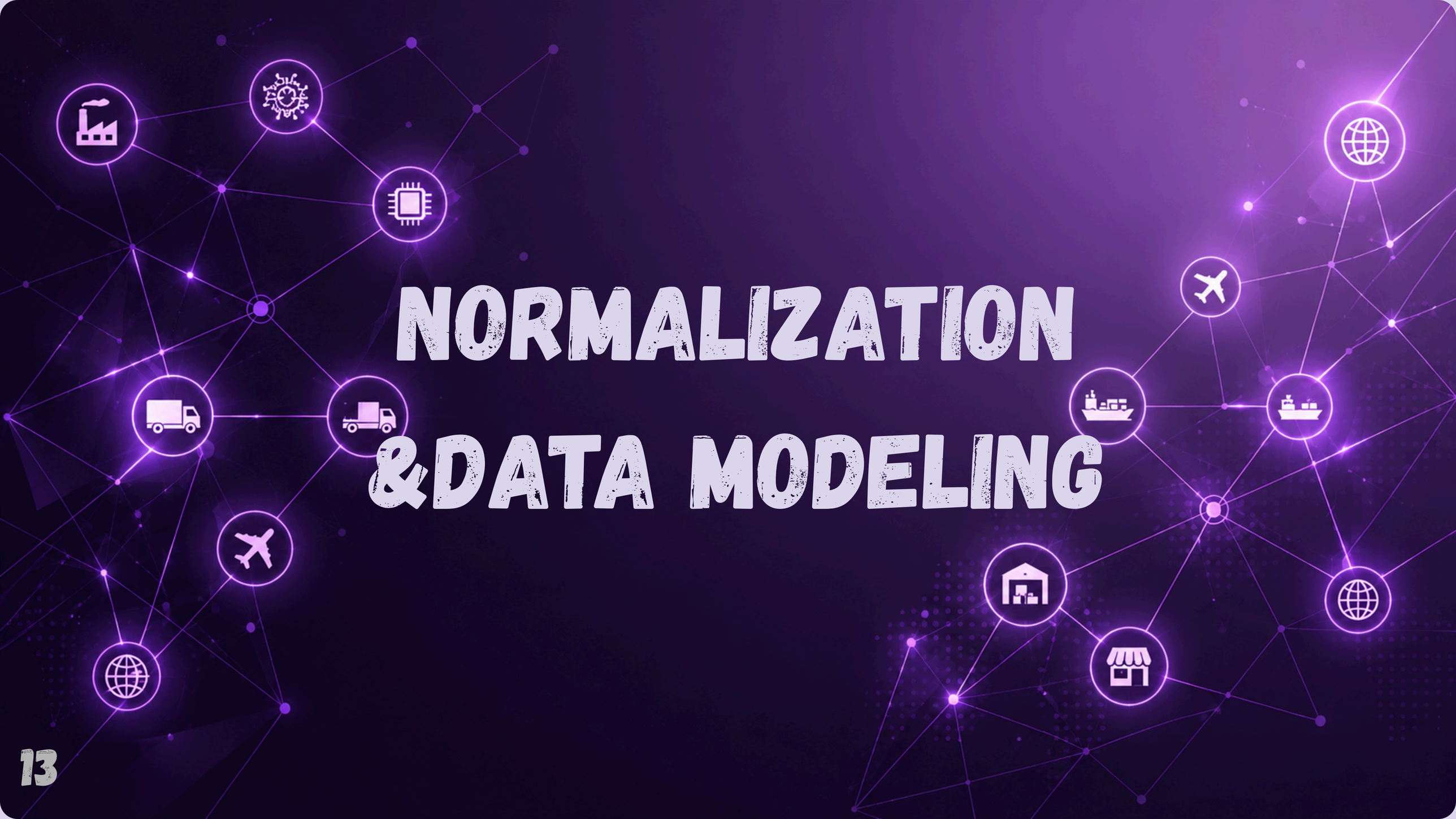
records verified for uniqueness

```
select *
from (
  select *,
   row_number() over (partition by Order_ID order by
  (select null)) as rn
  from electronics_supplyChain
) t
where rn = 1
```

Description: Compare records for exact or near matches. Remove copies to maintain a single source of truth.

DATA VALIDATION

```
# Remove Invalid Records
df = df[
    (df["Shipping times"] >= 0) &
    (df["Price"] >= 0) &
    (df["Discount (%)"].between(0,100)) &
    (df["Number of products sold"] > 0) &
    (df["Units Returned"] <= df["Number of products sold"])
]
```


NORMALIZATION
&DATA MODELING

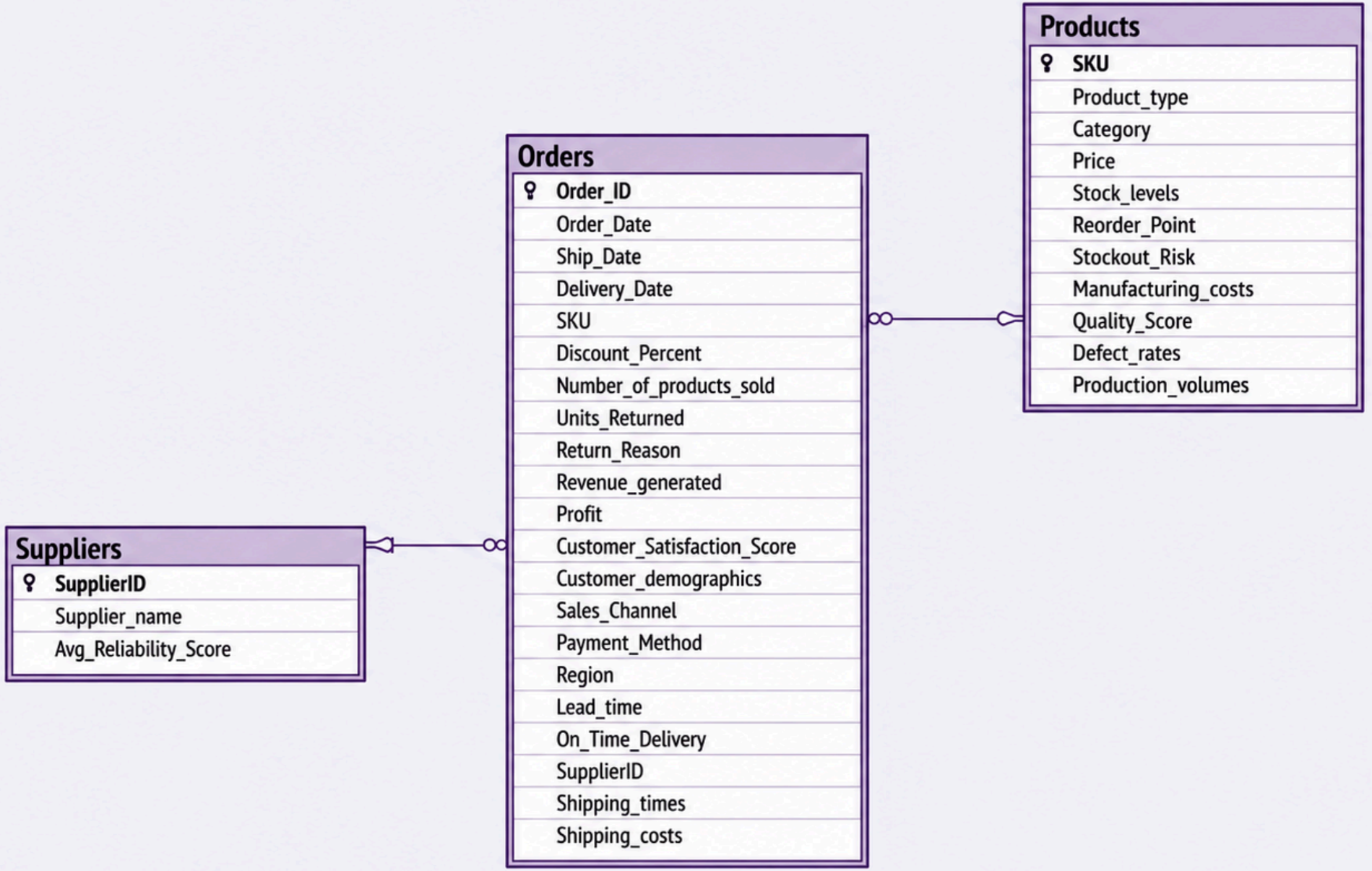
Normalization in SQL

Converting raw data into structured, clean, and relational tables.

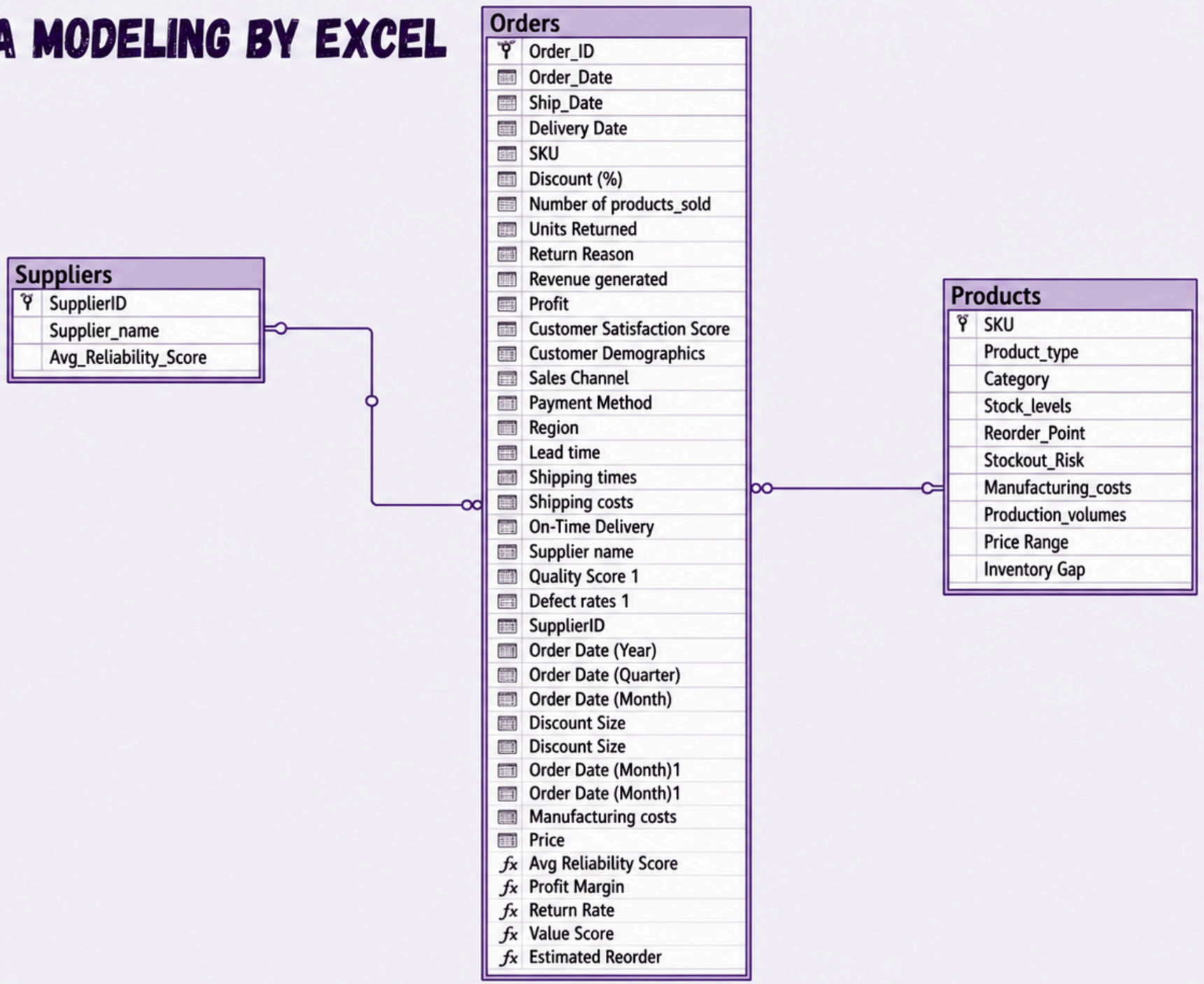
```sql
--Normalization

CREATE TABLE Suppliers (
    SupplierID INT PRIMARY KEY,
    Supplier_name VARCHAR(255),
    Avg_Reliability_Score FLOAT
)

CREATE TABLE Products (
    SKU VARCHAR(50) PRIMARY KEY,
    Product_type VARCHAR(100),
    Category VARCHAR(100),
    Price DECIMAL(10, 2),
    Stock_levels INT,
    Reorder_Point INT,
    Stockout_Risk VARCHAR(50),
    Manufacturing_costs DECIMAL(10, 2),
    Quality_Score FLOAT
    Defect_rates FLOAT,
    Production_volumes INT
)
CREATE TABLE Orders (
    Order_ID VARCHAR(50) PRIMARY KEY,
    Order_Date DATE,
    Ship_Date DATE,
    Delivery_Date DATE,
    SKU VARCHAR(50),
    Discount_Percent DECIMAL(5, 2),
    Number_of_products_sold INT,
    Units_Returned INT,
    Return_Reason VARCHAR(255),
    Revenue_generated DECIMAL(15, 2),
    Profit DECIMAL(15, 2),
    Customer_Satisfaction_Score FLOAT,
    Customer_demographics VARCHAR(100),
    Sales_Channel VARCHAR(100),
    Payment_Method VARCHAR(100),
    Region VARCHAR(100),
    Lead_time INT,
    On_Time_Delivery VARCHAR(50),
    SupplierID INT,
    Shipping_times decimal(30,2),
    Shipping_costs decimal(30,2),

    CONSTRAINT FK_Orders_Products FOREIGN KEY (SKU)
        REFERENCES Products(SKU),

    CONSTRAINT FK_Orders_Suppliers FOREIGN KEY (SupplierID)
        REFERENCES Suppliers(SupplierID)
)

SELECT * FROM electronics_supplyChain

WITH SupplierAgg AS (
  SELECT
     Supplier_name,
     AVG(Supplier_Reliability_Score) AS Avg_Reliability_Score
  FROM electronics_supplyChain
  GROUP BY Supplier_name
)
INSERT INTO Suppliers (SupplierID, Supplier_name, Avg_Reliability_Score)
SELECT
   DENSE_RANK() OVER (ORDER BY Supplier_name) AS SupplierID,
   Supplier_name,
   Avg_Reliability_Score
FROM SupplierAgg

INSERT INTO Products (
   SKU,Product_type,Category,Price,Stock_levels,Reorder_Point,Stockout_Risk,
   Manufacturing_costs,Quality_Score,Defect_rates,Production_volumes
)
SELECT
   SKU,Product_type,Category,Price,Stock_levels,Reorder_Point,Stockout_Risk,
   Manufacturing_costs,Quality_Score,Defect_rates,Production_volumes
FROM electronics_supplyChain

INSERT INTO Orders (
   Order_ID,Order_Date,Ship_Date,Delivery_Date,SKU,Discount_Percent,
   Number_of_products_sold,Units_Returned,Return_Reason,Revenue_generated,
   Profit,Customer_Satisfaction_Score,Customer_demographics,Sales_Channel,
   Payment_Method,Region,Lead_time,On_Time_Delivery,SupplierID,
   Shipping_times,Shipping_costs
)
SELECT
   Order_ID,Order_Date,Ship_Date,Delivery_Date,SKU,Discount_Percent,
   Number_of_products_sold,Units_Returned,Return_Reason,Revenue_generated,
   Profit,Customer_Satisfaction_Score,Customer_demographics,Sales_Channel,
   Payment_Method,Region,Lead_time,On_Time_Delivery,SupplierID,
   Shipping_times,Shipping_costs
FROM electronics_supplyChain
```

Description: This script normalizes the raw supply chain dataset into three relational tables: Suppliers, Products, and Orders.
It structures the data to improve consistency, eliminate redundancy, and support efficient analysis and reporting.

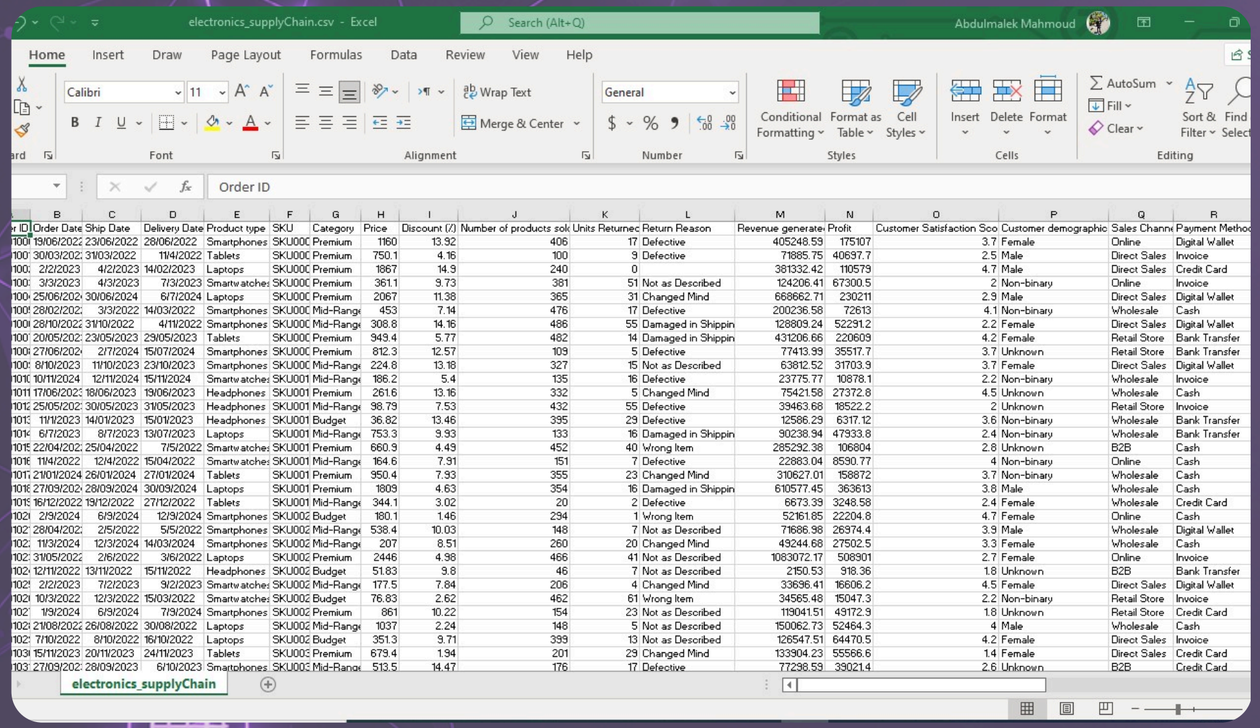
DATA MODELING BY SQL



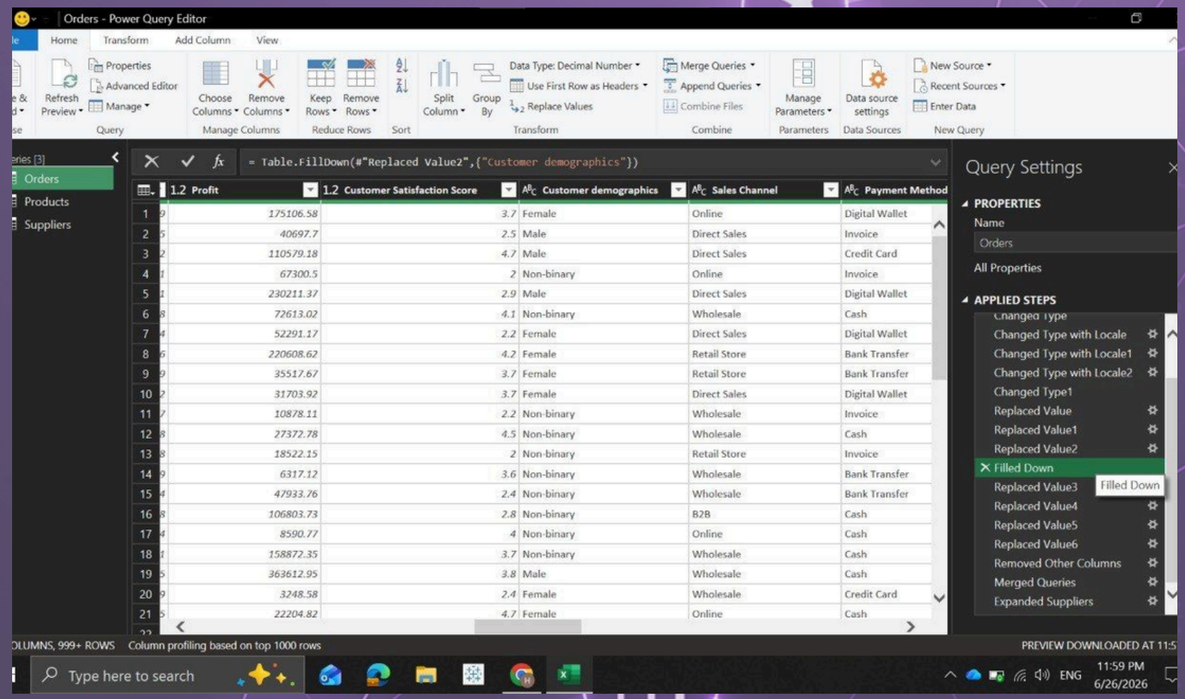
DATA MODELING BY EXCEL



ANALYSIS-READY DATASET



BEFORE



AFTER

Analysis-Ready Dataset

Order ID		Order Date		Ship Date		Delivery Date		SKU		Product Type	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%

SKU		Product Type		Category		Price		Stock Levels		Reorder Point	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%

Stock Levels		Reorder Point		Stockout Risk		Manufacturing Costs		Production Volumes		Shipping Times	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%

Payment Method		Region		Lead Time		Shipping	
Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%

Quality Score		Defect Rates		Supplier ID	
Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%

Profit		Customer Satisfaction Score		Customer Demographics		Sales Channel	
Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%

Discount (%)		Number of Products Sold		Units Returned		Return Reason		Revenue Generated	
Valid	100%	Valid	100%	Valid	100%	Valid	100%	Valid	100%
Error	0%	Error	0%	Error	0%	Error	0%	Error	0%
Empty	0%	Empty	0%	Empty	0%	Empty	0%	Empty	0%

DATA ANALYSIS (EDA)

WE HAVE SEGMENTED THE DATA INTO FIVE KEY DOMAINS FOR DETAILED ANALYSIS.

01

COMMERCIAL INSIGHTS DASHBOARD

A comprehensive view of sales performance, revenue trends, and growth opportunities across product lines.

MAXIMIZE PROFITABILITY & SALES REACH.

02

INVENTORY DASHBOARD

Real-time tracking of stock levels, inventory turnover rates, and optimization of supply-demand balance.

EFFICIENT STOCK MANAGEMENT.

03

CUSTOMER EXPERIENCE & RETURN

Analyzing feedback metrics, CSAT scores, and return patterns to enhance customer satisfaction.

ENHANCE CUSTOMER JOURNEY.

04

SUPPLIER PERFORMANCE

Evaluate supplier reliability, on-time delivery rates, and quality scores for a resilient supply chain.

ENSURE SUPPLY CHAIN STRENGTH.

05

SHIPPING ANALYTICS

Optimize delivery routes, minimize transit times, and analyze shipping costs for operational efficiency.

OPTIMIZE LOGISTICS & DELIVERY.

Commercial Insights Dashboard (Overview)

DOMAIN 1

- What is the total revenue?
- What is the total profit?
- What is the profit margin?
- What is the average discount percentage?
- How have revenue and profit changed quarterly?
- Which product generates the highest revenue and profit?
- Which sales channel generates the highest profit?
- Which region has the highest profit margin?
- How are discounts distributed?
- How do revenue and profit vary by customer gender?

Discount Distribution %

01

PYTHON

Code and Data Output

```python
df_clean['Discount Category'] = pd.cut(
    df_clean['Discount (%)'],
    bins=[0, 5, 10, 15],
    labels=['Low', 'Medium', 'High'],
    include_lowest=True
)

discount_distribution = (
    .value_counts(normalize=True)
    .sort_index()
    .null(10)
    .round(2)
    .reset_index()
)
discount_distribution.columns = [
    "Discount Category",
    "Percentage (%)"
]

discount_distribution
```

	Discount Category	Percentage (%)
0	Low	33.2
1	Medium	31.4
2	High	35.4

02

SQL

Query and Data Output

```sql
WITH DiscountCategories AS (
      SELECT
            WHEN Discount >= 0 AND Discount <= 5 THEN 'Low'
            WHEN Discount > 5 AND Discount <= 10 THEN 'Medium'
            WHEN Discount > 10 AND Discount <= 15 THEN 'High'
      END AS Discount_Category
   FROM dbo.electronics_supplyChain
   WHERE Discount IS NOT NULL
)
SELECT
   Discount_Category,
   ROUND(
      COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(),
      2
   ) AS [Percentage (%)]
) AS [Percentage (%)]
FROM DiscountCategories
GROUP BY Discount_Category
ORDER BY
   CASE Discount_Category
      WHEN 'Low' THEN 1
      WHEN 'Medium' THEN 2
      WHEN 'High' THEN 3
   END;
```

Results Messages

	Discount_Categ	Percentage (%)
1	Low	33.2000000000
2	Medium	31.4000000000
3	High	35.4000000000

Inventory Management Dashboard (Overview)

DOMAIN 2

- What is the production volume?
- What are the total units sold?
- What is the total stock level?
- What is the size of the inventory gap?
- How does production volume vary by product type?
- Which SKUs are the 5 top and bottom by units sold?
- What is the relationship between price range and profit margin?
- How do defect rates compare across product types?
- Which product types are most at risk of stockout?
- How do product types rank by inventory gap versus stock levels?

SKU Analysis

01

SQL QUERY AND RESULTS

Code and Data Output

```sql
-- Chart 1: SKUs by Stockout Risk
SELECT [Stockout_Risk] AS stockout_risk,
  COUNT(*) AS sku_count,
  CAST(COUNT(*) * 100.0 /
  SUM(COUNT(*)) OVER() AS DECIMAL(5, 1)) AS percentage
FROM electronics_supplyChain
GROUP BY [Stockout_Risk]
ORDER BY sku_count DESC;
```

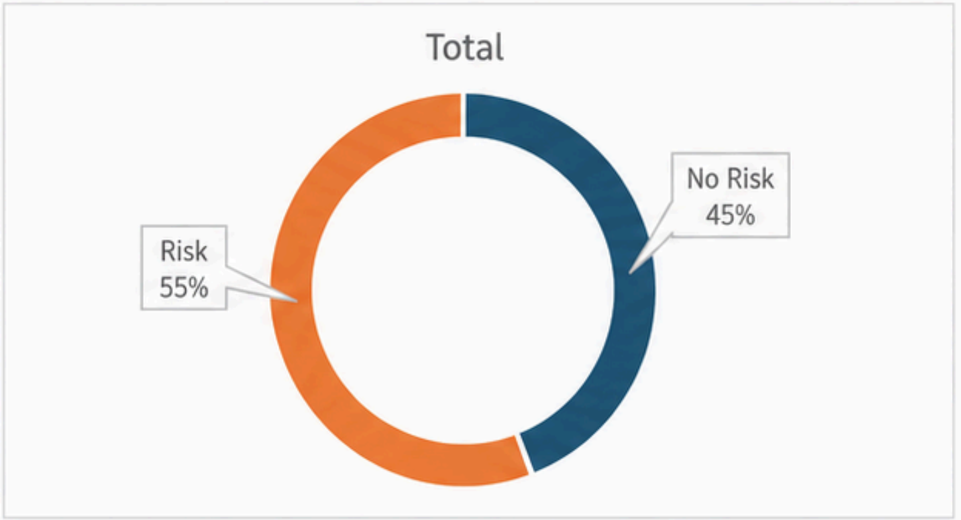
stockout_risk	sku_count	percentage
1	831	55.4
0	669	44.6

02

EXCEL PIVOT AND CHART

Pivot Table and Donut Visualization

Row Labels	Count of SKU
No Risk	44.60%
Risk	55.40%
Grand Total	**100.00%**



Customer Experience & Return Dashboard (Overview)

DOMAIN 3

- What is the average return rate?
- Which region has the highest return rate and highest return loss?
- What is the total units returned?
- Which sales channel has the highest return rate?
- What is the average satisfaction score?
- Which product type has the highest number of units returned?
- What is the total estimated return loss?
- How are product returns distributed across different regions?
- How do average satisfaction scores vary by customer gender?
- How have return rate, unit count, and loss metrics changed compared to the previous year (vs PY)?

Average satisfaction score for each gender

01

PYTHON

Code and Data Output

```
gender_satisfaction = (
    df_clean
    .groupby("Customer demographics")["Customer Satisfaction Score"]
    .mean()
    .sort_values(ascending=False)
    .reset_index()
)

gender_satisfaction
```

✓ 0.0s

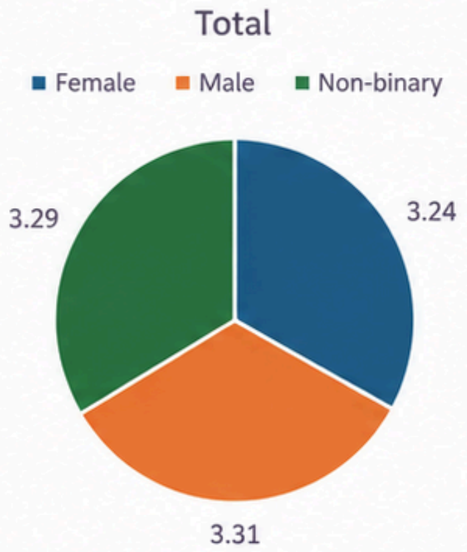
	Customer demographics	Customer Satisfaction Score
0	male	3.308083
1	non-binary	3.292484
2	female	3.237628

02

EXCEL

Pivot Table and Chart Output

Row Labels	Average of Customer Satisfaction Score
Female	3.24
Male	3.31
Non-binary	3.29
Grand Total	**3.28**



Suppliers Performance Dashboard (Overview)

DOMAIN 4

- What is the overall average quality score for all suppliers?
- What is the average reliability score for the supply chain?
- What is the average defect rate across all products?
- How does quality score compare to manufacture cost across different products?
- Which supplier generates the highest profit and has the best quality?
- Which supplier has the best value score?
- Which supplier has the highest average return rate?
- Which supplier has the highest defect rate?
- How have supplier return rates changed over time?
- How do key metrics like quality and reliability compare to previous year (PY) performance?

Supplier Value Score

01

PYTHON

Code and Data Output

```python
supplier_value_score = (
    df_clean.groupby("Supplier name")
    .agg({
        'Quality Score': 'mean',
        'Supplier Reliability Score': 'mean',
        'Defect rates': 'mean'
    })
)
supplier_value_score["Value Score"] = (
    supplier_value_score["Supplier Reliability Score"] /
    supplier_value_score["Defect rates"]
)
supplier_value_score = (
    supplier_value_score[["Value Score"]]
    .round(3)
    .sort_values(by='Value Score', ascending=False)
    .reset_index()
)
supplier_value_score
```

	Supplier name	Value Score
0	supplier 4	7.038
1	supplier 1	6.907
2	supplier 3	6.796
3	supplier 2	6.408
4	supplier 5	5.902

02

SQL QUERY

Query and Results Output

```sql
SELECT
    s.Supplier_name,
    ROUND(
        (AVG(p.Quality_Score) * AVG(s.Avg_Reliability_Score))
        / AVG(p.Defect_rates),
        3
    ) AS Value_Score
FROM Orders o
JOIN Suppliers s
    ON o.SupplierID = s.SupplierID
JOIN Products p
    ON o.SKU = p.SKU
GROUP BY s.Supplier_name
ORDER BY Value_Score DESC;
```

	Supplier_name	Value_Score
1	supplier 4	7.039
2	supplier 1	6.903
3	supplier 3	6.795
4	supplier 2	6.407
5	supplier 5	5.903

Shipping Analytics Insights (Overview)

DOMAIN 5

- What is the total shipping cost?
- What is the average lead time?
- What is the average ship time?
- What is the average shipping cost?
- Which region has the highest number of units and shipping cost?
- How does shipping performance vary by product type?
- Which product type has the highest number of units sold?
- How many orders are on-time versus delayed?
- What is the distribution of shipping performance across regions?
- How do average shipping days differ by region?
- What is the profit associated with different delivery statuses?
- What is the profit associated with different delivery statuses?

Shipping Time per Region

01

SQL Query

```
766  -- Insight: Shipping Time per Region
767     Region AS Regions,
768     CAST(AVG(Shipping_Times) AS DECIMAL(10, 2)) AS avg_shipping_time
769     FROM electronics_supplyChain
770     GROUP BY Region
771     ORDER BY Regions ASC;
776
```

89 %

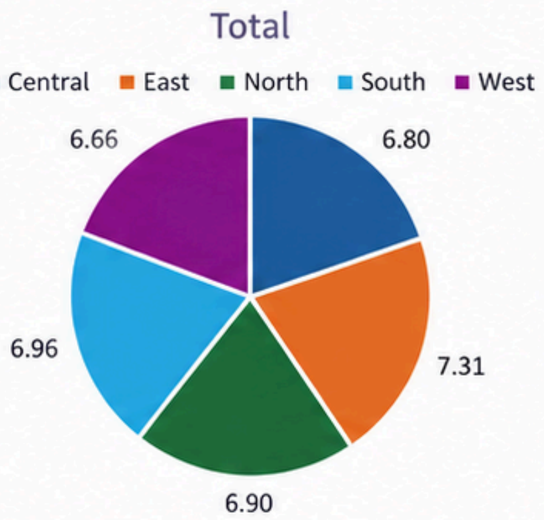
Results | Messages

	Regions	avg_shipping_time
1	central	6.80
2	east	7.31
3	north	6.90
4	south	6.96
5	west	6.66

02

Excel Pivot Table & Chart

Row Labels	Average of Shipping times
Central	6.80
East	7.31
North	6.90
South	6.96
West	6.66
Grand Total	**6.93**



VISUALIZATION

Region (All) | Payment (All) | 2022 2023 2024 | headphones laptops smartphones smartwatch.. tablets

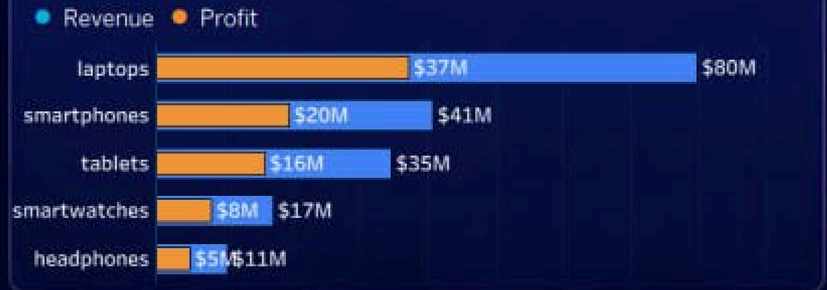
Overview | Inventory | Returns & CSAT | Suppliers | Shipping

Total Revenue
$183M
▲ 2.1% vs PY

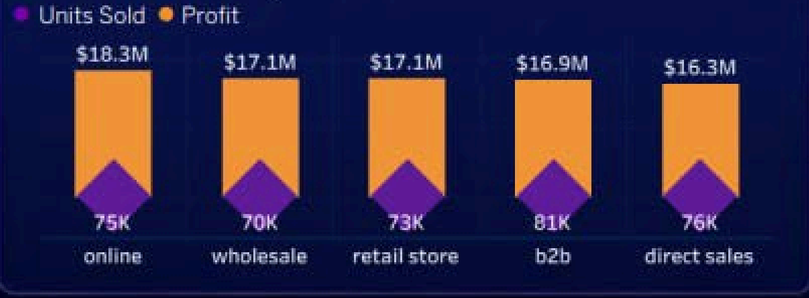
Total Profit
$86M
▼- 3.2% vs PY

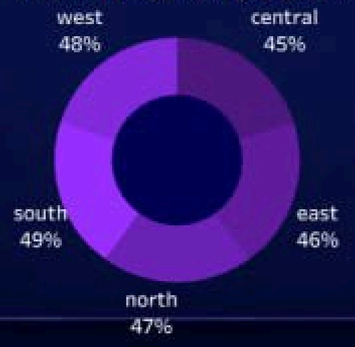
Profit Margin %
46.8%
▼-0.5%PP vs PY

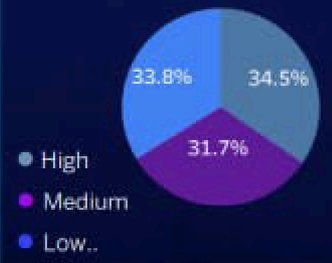
AVG Discount %
7.5%
▲ 6.1% PP vs PY

Revenue & Profit By Product


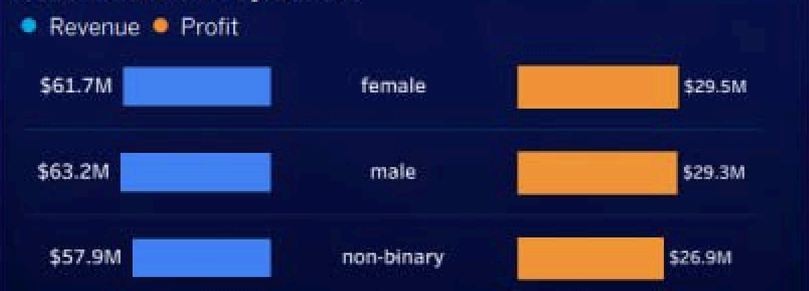
Revenue & Profit Trend (Quarterly)

Profit & Units Sold By Sales Channel


Profit Margin % By Region


Discount % Distribution (by Units Sold)


Revenue & Profit By Discount

Revenue & Profit By Gender


Activate Windows
Go to Settings to activate Windows.

Commercial Insights Dashboard | Overview

Region: All

Year: 2022 | 2023 | 2024

Product type: headphones | laptops | smartphones | smartwatche | tablets

Overview | Inventory | CSAT & RETURNS | Supplier | Shipping

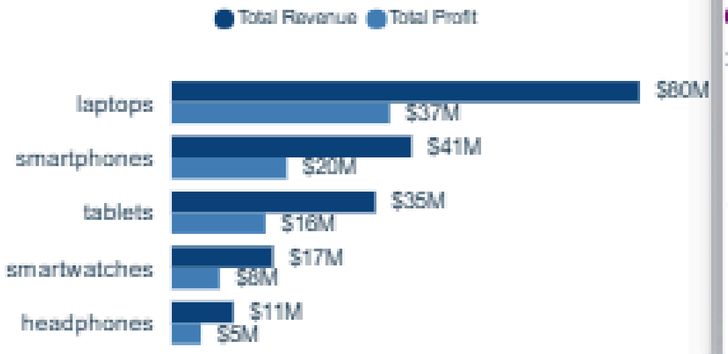
Total Revenue
$183M

Profit Margin
46.8%

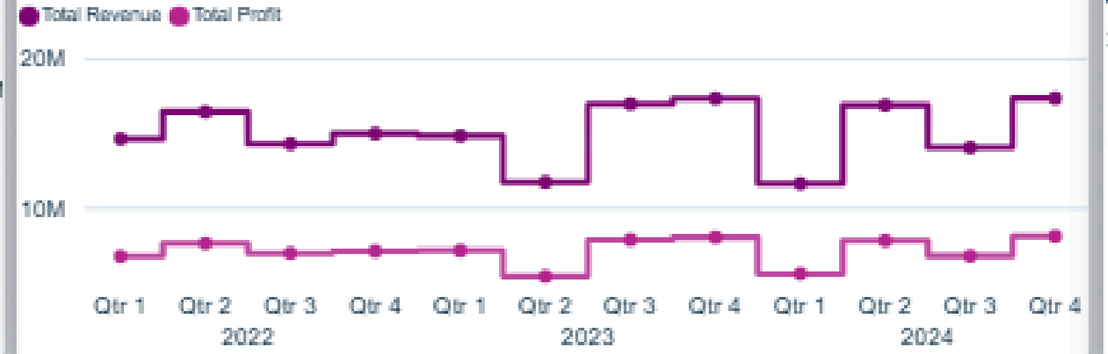
Avg Discount
7.5

Total Profit
86M

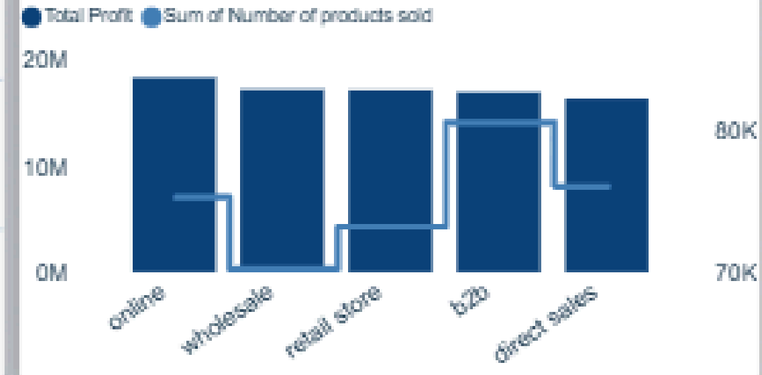
Revenue & Profit by Product Type



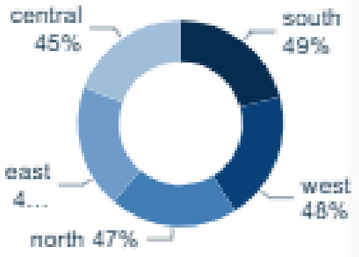
Revenue & Profit Trend (Quarterly)



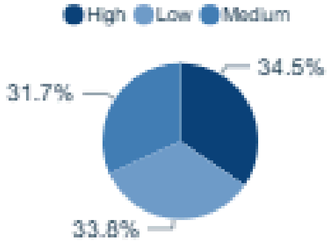
Profit & Units Sold By Sales Channel



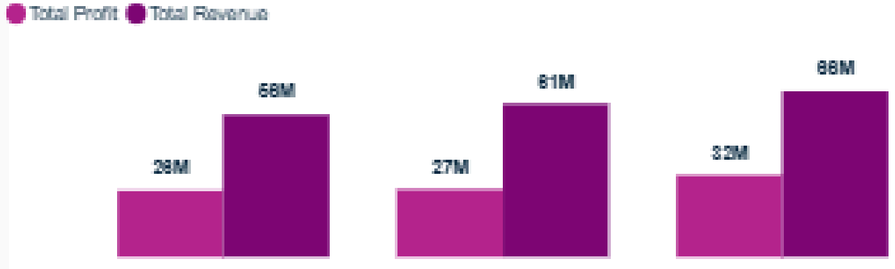
Profit Margin by Region



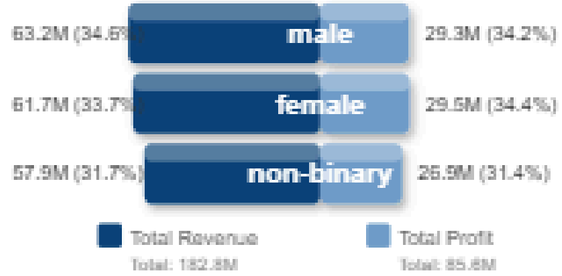
Discount Distribution (by unit sold)



Total Profit and Total Revenue by Discount Range



Total Revenue & Total Profit by Customer demographics



COMMERCIAL INSIGHTS

01

1. REVENUE VS. PROFIT

- Revenue grew by **2.1%** to **$182.81M**, but total profit fell by **3.2%** to **$85.63M**, signaling an urgent need to optimize operational costs.

KEY TAKEAWAY: Focus on controlling costs to protect profit.

02

2. PRODUCT LEADERSHIP

- Laptops are the primary revenue driver (**$79.7M**) and profit contributor (**$37.1M**).

KEY DRIVER: Laptops lead in both revenue and profit.

03

3. SALES CHANNELS

- Online sales outperform other channels in both volume and profit.

CHANNEL INSIGHT: Online is the most effective sales channel.

04

4. MARKET STABILITY

- Profit margins remain steady across all regions (**45%–49%**) and are consistent across all gender demographics.

STABILITY FACTOR: Strong and consistent margin performance.

05

5. DISCOUNT IMPACT

- The current discount strategy requires review, as higher discount levels do not guarantee proportional profit growth.

ACTION POINT: Reassess discount strategy to improve profitability.

Stockout Risk: (All) | Category: (All) | Product type: (All) | Region: (All) | 2022 | 2023 | 2024

Overview | Inventory | Returns & CSAT | Suppliers | Shipping

Production Volume
894K
▼-0.01% vs PY

Total Units Sold
375K
▲ 9.0% vs PY

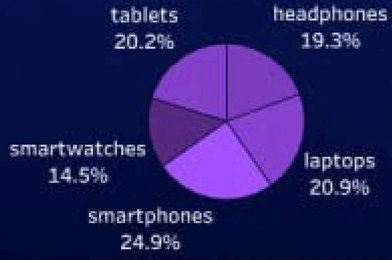
Total Stock
231K
▲ 5.1% vs PY

Inventory Gap
288K
▼- 0.2 vs PY

Products
1,500
▲ 6.7% vs PY

%Stockout Risk

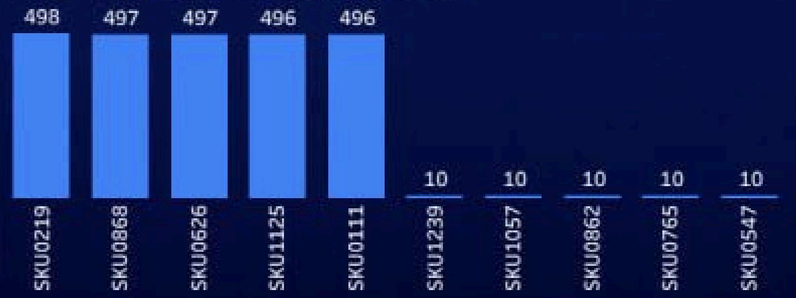
%Production Volumes By Product Type



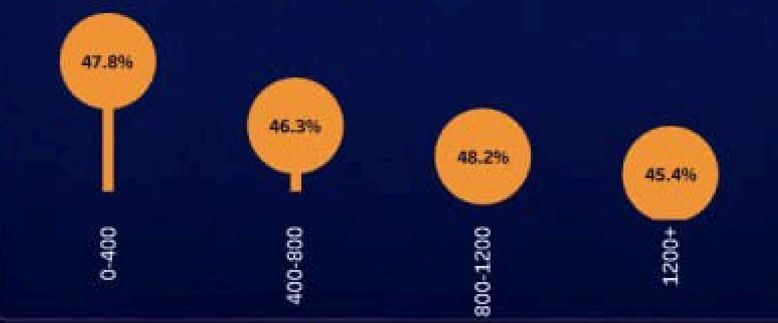
Profit Margin Vs Unit Sold By Products

Inventory Gap & Stock Level By Product Type

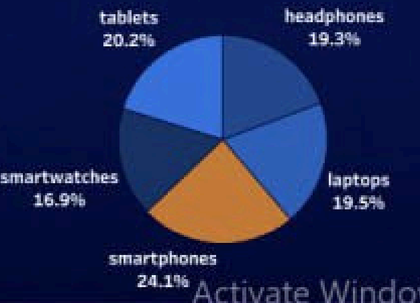
Top VS Bottom 5 SKUs By Unit Sold



SKUs By Price Range & Profit Margin



%Defect Rates By Product Type



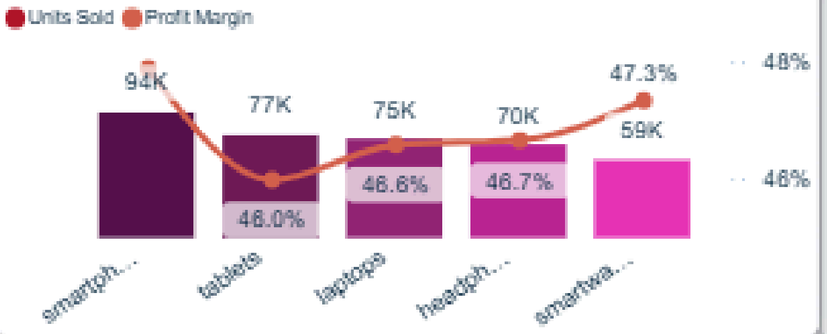
Activate Windows
Go to Settings to activate Windows.

Stockout Risk: All | Product type: All | Region: All | Year: 2022 2023 2024

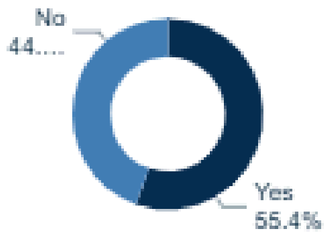
Overview | Inventory | CSAT & RETURNS | Supplier | Shipping

Production Volume	Sold Units	Stock Level	Inventory Gap	#Products
894K	375K	231K	288K	1.5K

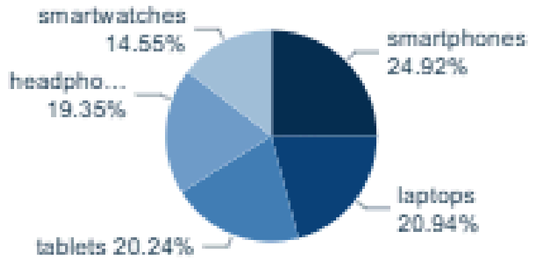
Customer Satisfaction & Units Returned per Sales Channel



%Stockout Risk

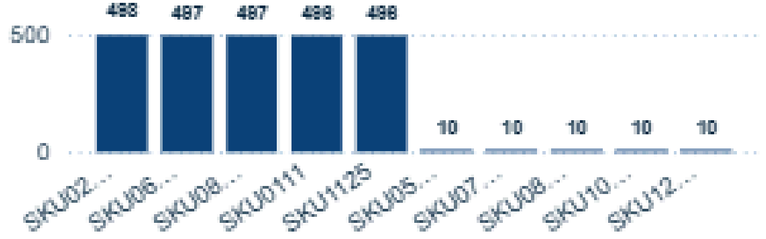


Production Volume by Product Type

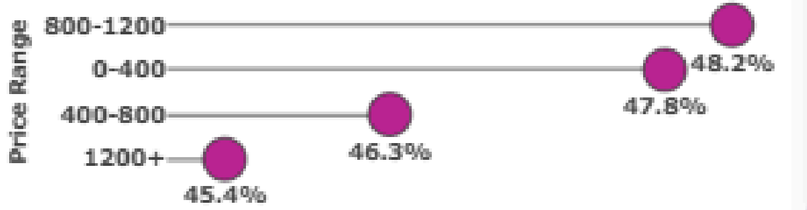


Inventory Gap and Sum of Stock levels by Product type

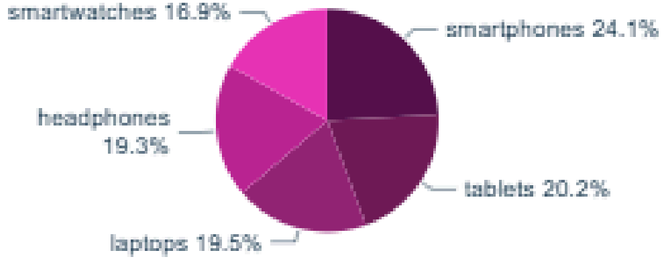
Sum of Units by SKU



Profit Margin by Price Range



%Defect Rate by Product Type



INVENTORY

01

1. STRONG SALES GROWTH

- Total Units Sold hit **375K (+9.0% YoY),** driven by **Smartphones** leading at **94.3K units.**

DRIVER: Smartphones dominate volume.

02

2. HIGH STOCKOUT RISK

- A critical **55.4% of SKUs** are at risk of **running out**, leaving only **44.6%** secure.

RISK ASSESSMENT: Over half of SKUs at risk.

03

3. SUPPLY DISCONNECT

- Smartphones hold the highest stock **(55.8K)** but face the largest inventory gap **(72.8K)**, showing a lag in meeting demand.

ALIGNMENT ISSUE: Smartphone supply lags demand.

04

4. PREMIUM MARGIN WINS

- **The $800–$1,200** price tier drives the highest profitability with a **48.16%** profit margin.

PROFIT TIER: Premium drives high margins.

05

5. QUALITY CONTROL ISSUE

- **Smartphones** register the highest defect rate at **24.1%**, requiring immediate attention.

URGENT ACTION: Defect rate high in Smarphones.

Region (All) | Return Reason (All) | Customer demo (All) | 2022 | 2023 | 2024

Overview | Inventory | Returns & CSAT | Suppliers | Shipping

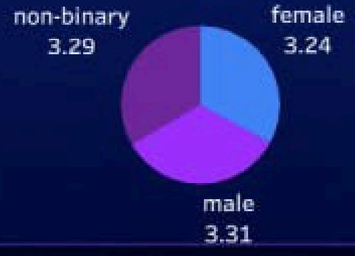
AVG. Return Rate
7.4%
▲ 2.1% vs PY

Total Units Returned
27.6K
▲ 11.3% vs PY

Total Estimated Return Lose
$13M
▲ 9.7% vs PY

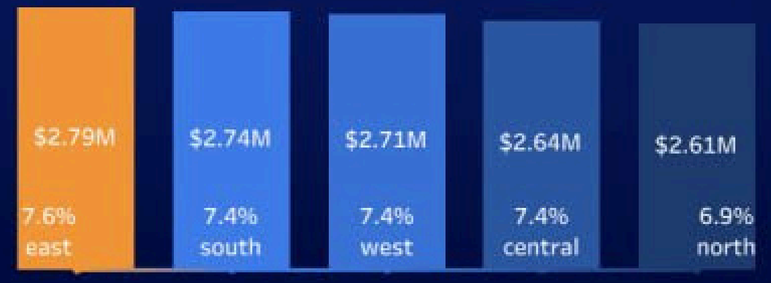
AVG. satisfaction score
3.28/5
▲ 0.0%
the same as the PY

AVG Satisfaction Score By Each Gender



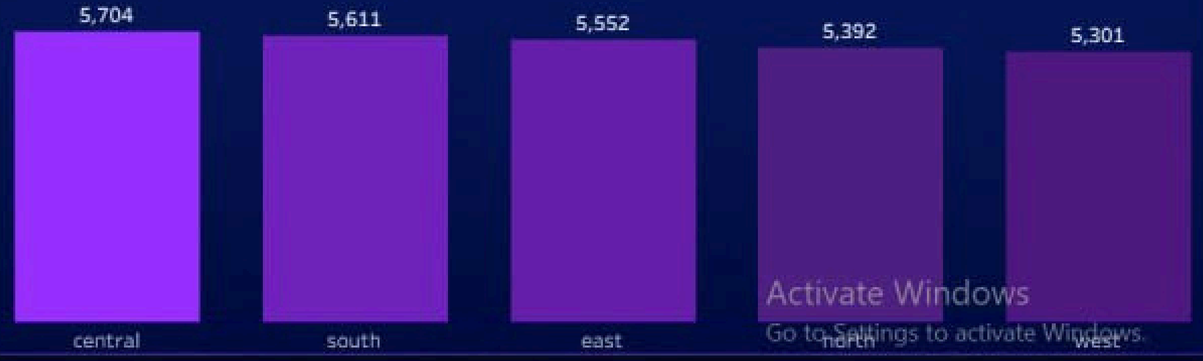
Customer Satisfaction & Units Returned By Sales Channel

%Return Rate VS Return Loss By Region



Units Returned By product type

Number of Product Return By Region



Activate Windows
Go to Settings to activate Windows.

Stockout Risk: All | Product type: All | Region: All | Year: All

Overview | Inventory | CSAT & RETURNS | Supplier | Shipping

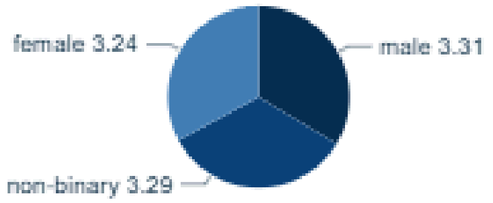
Avg Return Rate
7.35%

Returned Units
27.6K

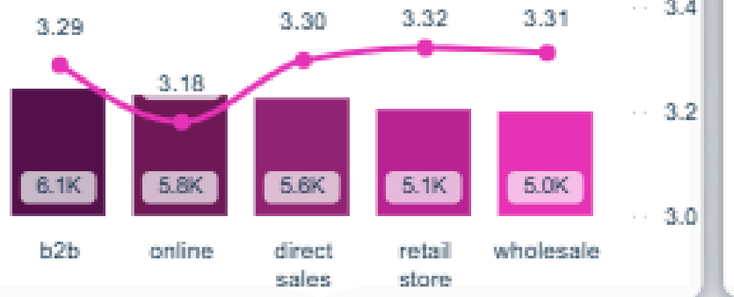
Estimated Return Loss
14M

Avg Return Rate
3.28/5

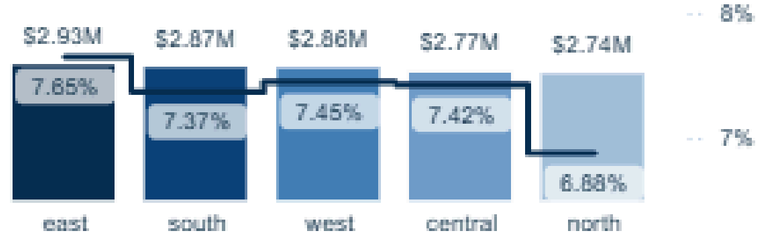
Avg Satisfaction Score per Gender



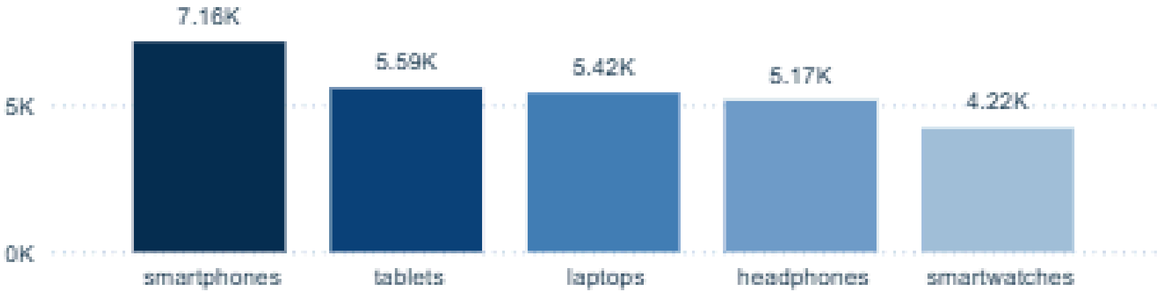
Customer Satisfaction & Units Returned per Sales Channel



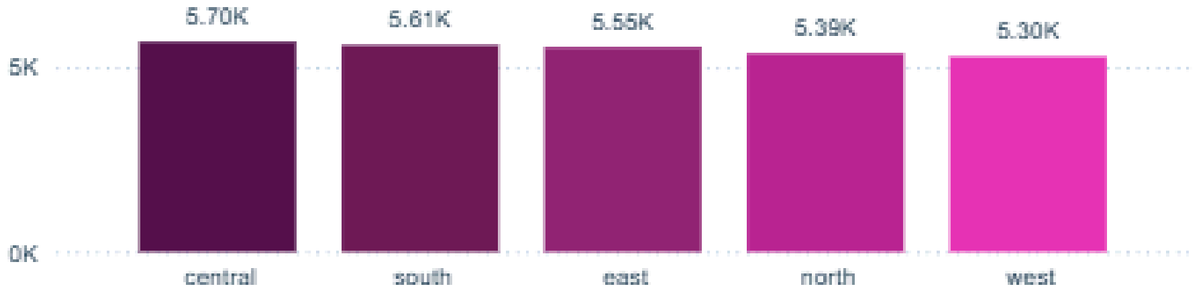
Return Loss and Avg Return Rate by Region



Sum of Units Returned by Product type



Sum of Units Returned by Region



CUSTOMER EXPERIENCE & RETURN

01

1. FINANCIAL IMPACT

- Total return losses reached **$13.50M (+9.7% vs PY),** driven by an increased return rate of **7.38%.**

INCREASED LOSSES: Driven by higher return rate.

02

2. VOLUME SURGE

- Total returned items jumped to **27,560 units (+11.3% vs PY),** while Customer Satisfaction (CSAT) remained completely stagnant at **3.280.**

UNITS UP, CSAT FLAT: Significant volume increase, no customer satisfaction gain.

03

3. PRIMARY DRIVERS

- * **Top Product:** Smartphones (7,157 units).
- * **Top Channel:** B2B (6,069 units) & Online (5,788 units).
- * **Top Region:** East Region ($2.79M loss) & Central Region (5,704 units).

DRIVING FACTORS: Key product, channel, and regional trends identified.

04.

4. STRATEGIC FOCUS

- Urgent intervention is required to address key **systemic issues.** A critical, data-driven plan is being deployed.

OPTIMIZE QA: Optimize QA for high-return items.

AUDIT POLICIES: Audit return policies for clarity and impact.

ADDRESS INEFFICIENCIES: Streamline regional operations to remove bottlenecks.

05.

5. POLICY REVIEW & AUDIT

- *Audit **B2B & Online** return policies to reduce friction and cost.
- *Address systemic inefficiencies in East & Central regional operations.

ACTIONABLE STEPS: Specific areas for process improvement.

Supplier name: (All)

Region: (All)

2022 2023 2024

Overview | Inventory | Returns & CSAT | Suppliers | Shipping

AVG Quality Score

50.2%

▲ 3.1% vs PY

%AVG Reliability Score

65.3%

▲0.07% vs PY

%AVG Return Rate

7.4%

▲ 2.1% vs PY

%AVG Defect Rate

5%

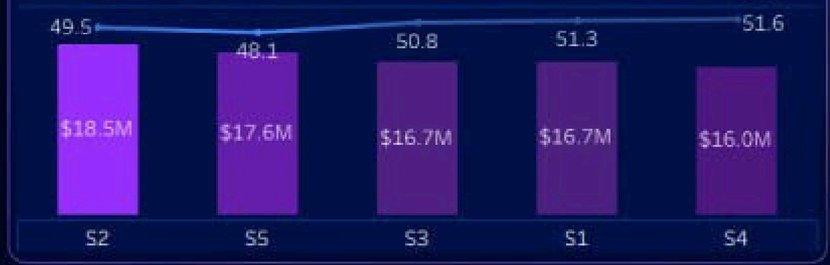
▼- 0.1 vs PY

Quality VS Manufacture cost

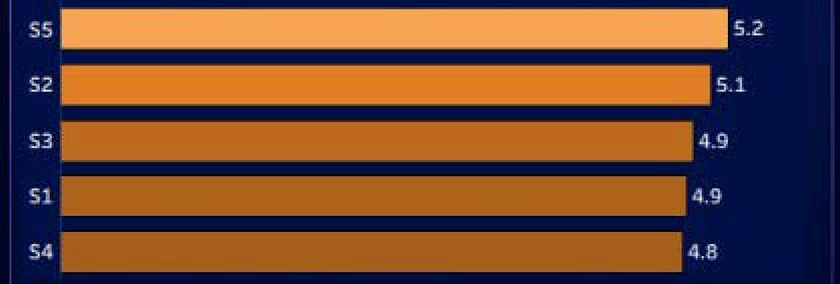
S1	Avg. Manufacturing costs	$214
	Avg. Quality Score	51.3%
S2	Avg. Manufacturing costs	$236
	Avg. Quality Score	49.5%
S3	Avg. Manufacturing costs	$231
	Avg. Quality Score	50.8%
S4	Avg. Manufacturing costs	$219
	Avg. Quality Score	51.6%
S5	Avg. Manufacturing costs	$210
	Avg. Quality Score	48.1%

Supplier Value Score

Supplier Profit & Quality



Defect Rate By Supplier



AVG Return Rate By Supplier

	2022	2023	2024
S1	6.57	8.15	8.00
S2	8.09	6.49	7.03
S3	7.28	7.44	6.98
S4	7.42	7.27	7.31
S5	7.52	7.65	8.00

Activate Windows
Go to Settings to activate Windows.

Product type: All

Region: All

Year: 2022 | 2023 | 2024

Overview | Inventory | CSAT & RETURNS | Supplier | Shipping

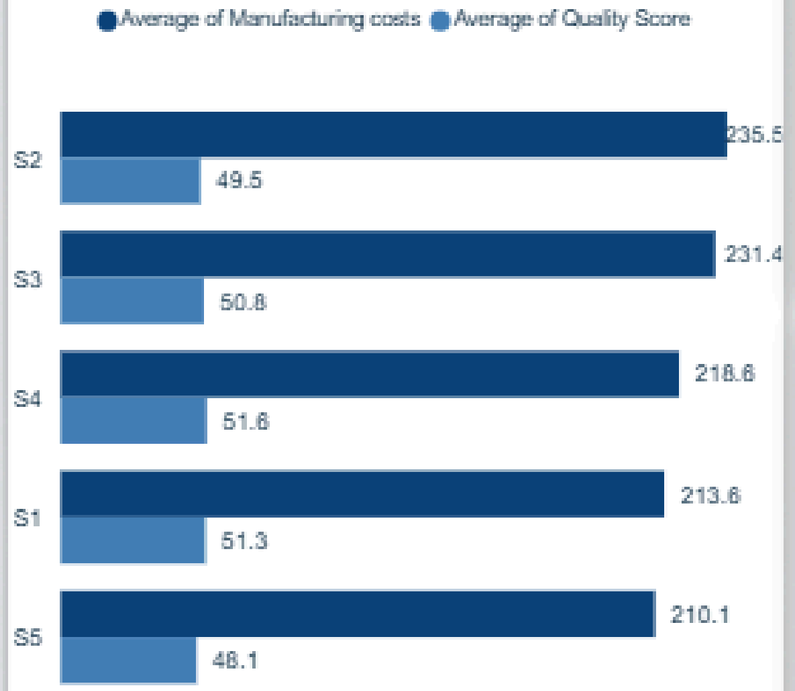
Avg Quality Score 50.2%

Avg Reliability Score 65.3%

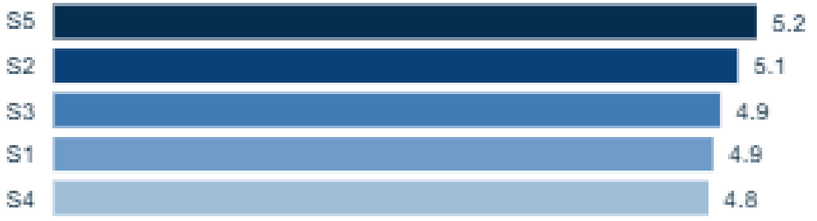
Avg Return Rate 7.3%

Avg Defect Rate 5%

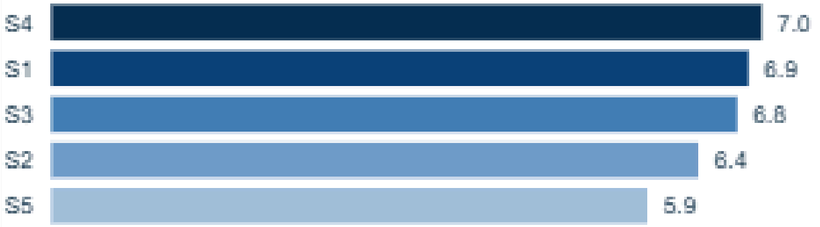
Quality VS Manufacturing Cost



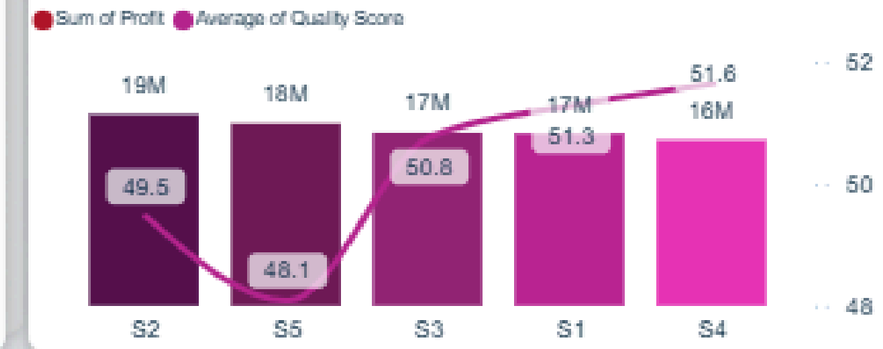
Average of Defect rates by Supplier Short



Supplier Value Score by Supplier Short



Supplier Profit& Quality



Avg Return Rate per Supplier

Supplier Short	2022	2023	2024
S1	6.82%	8.16%	8.20%
S2	7.92%	6.03%	6.89%
S3	7.07%	7.46%	6.88%
S4	7.23%	7.04%	6.87%
S5	7.75%	7.80%	8.44%

SUPPLIER PERFORMANCE

01

1. GLOBAL KPIs

- Average Quality Score stands at 50.23 (+3.1% vs PY). Defect Rate improved slightly to 4.98% (-0.1% vs PY, to 4.9%.
- Return Rate remains high at 7.38% (+2.1% vs PY).

MIXED TRENDS: Quality is up, but returns are rising.

02

2. TOP PERFORMER (Supplier 4)

- Achieved the highest **Value Score** (7.039). Recorded highest Quality
- Recorded highest **lowest Defect Rate** (4.84%).

RELIABLE SUPPLIER: Leading operationally.

03

3. UNDERPERFORMER (Supplier 5)

- Ranks lowest across critical metrics.
- **Worst Defect Rate** (5.19%).
- **Lowest Quality** (4.81) and lowest overall **Value Score** (5.903).

ACTION REQUIRED: Failing metrics.

04

4. PROFIT vs. QUALITY PARADOX

- Supplier 2 has highest profit ($18.5M) but high costs (235.5) & average quality.
- Highest-quality supplier (Supplier 4) has lowest profit ($16.0M).

DISCONNECT: Highest profit isn't highest quality.

05

5. STRATEGIC ACTION

- Restructure or issue a strict quality mandate for Supplier 5.
- Renegotiate terms with Supplier 4.

INTERVENTION NEEDED: Mandates and negotiation.

On-Time vs Delayed: (All)

Region: (All)

2022 2023 2024

Overview | Inventory | Returns & CSAT | Suppliers | Shipping

Total Shipping Cost
$262.4K
▲ 6.7% vs PY

AVG Shipping Cost
$176
▲ 0.7% vs PY

AVG_Ship_Days
6.9
▼ -0.04% vs PY

AVG_Lead_Days
13.5
▼ -0.02% vs PY

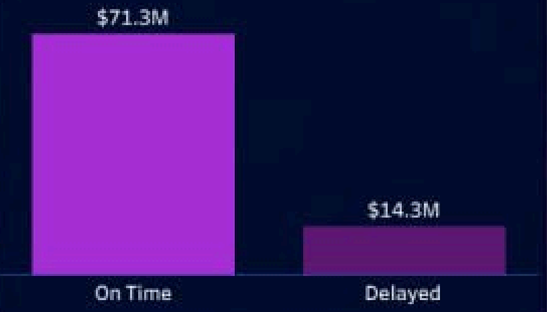
Region Shipping Performance

north	Units	78.4K
	Ship cost	$51.0K
central	Units	76.8K
	Ship cost	$55.6K
south	Units	76.2K
	Ship cost	$54.4K
east	Units	72.6K
	Ship cost	$53.8K
west	Units	71.2K
	Ship cost	$47.6K

Order Delivery Status
(# Orders & % Delivery status)

AVG Shipping Days By Region

Profit by Delivery Status



Shipping Performeance: Units & Total Cost By Product Type

Delivery Performance ByRegion (# Orders)

north	248 On Time	65 Delayed
south	255 On Time	57 Delayed
east	230 On Time	67 Delayed
central	241 On Time	49 Delayed
west	232 On Time	56 Delayed

Activate Windows
Go to Settings to activate Windows.

On-Time Delivery: All

Region: All

Year: 2022 | 2023 | 2024

Overview | Inventory | CSAT & RETURNS | Supplier | Shipping

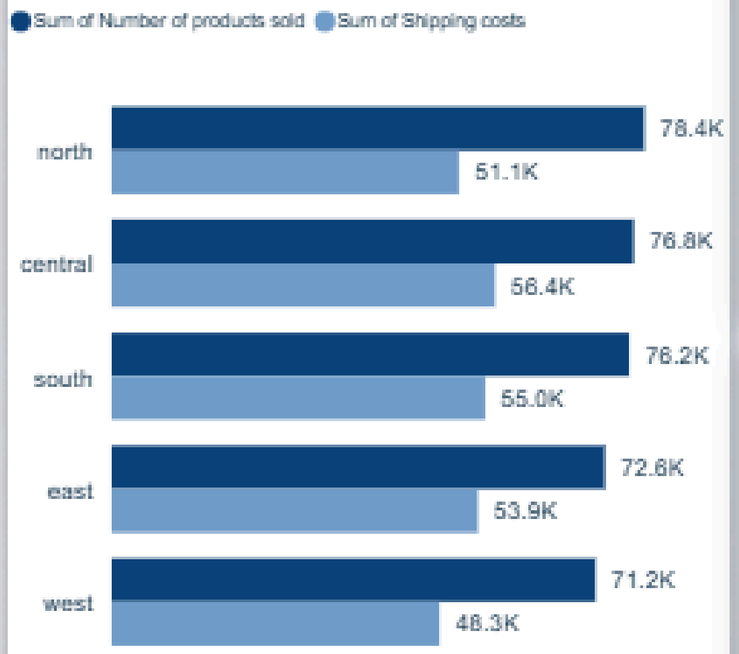
Shipping Cost 265K

Avg Ship Cost $176

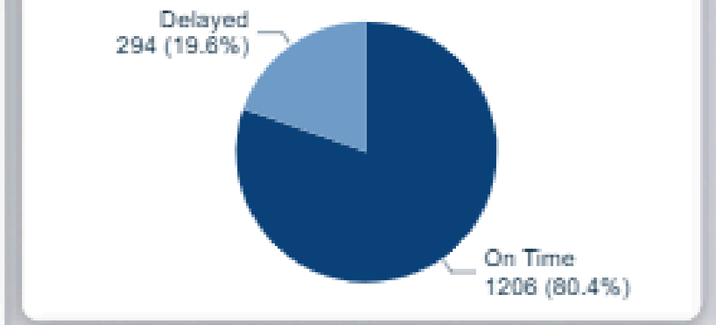
Avg Ship Time 6.9

Avg Lead Time 13.5

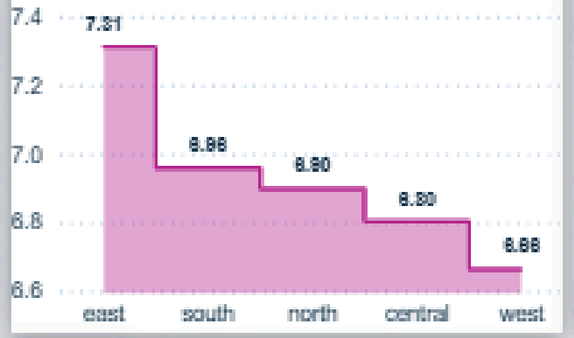
Region Shipping Performance



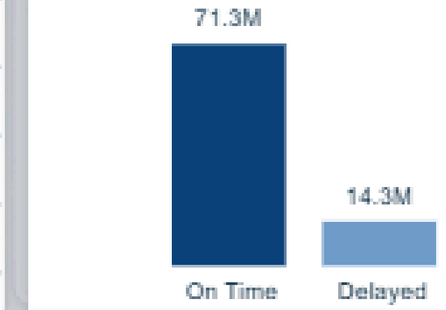
Order Delivery Status



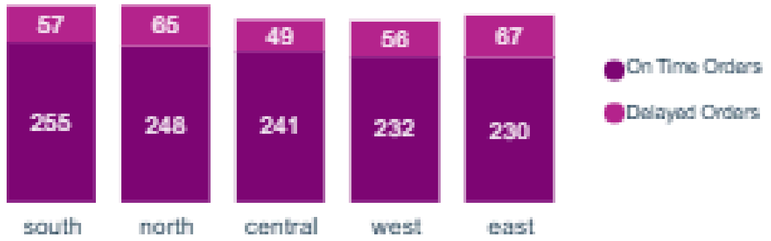
Avg Shipping Time by Region



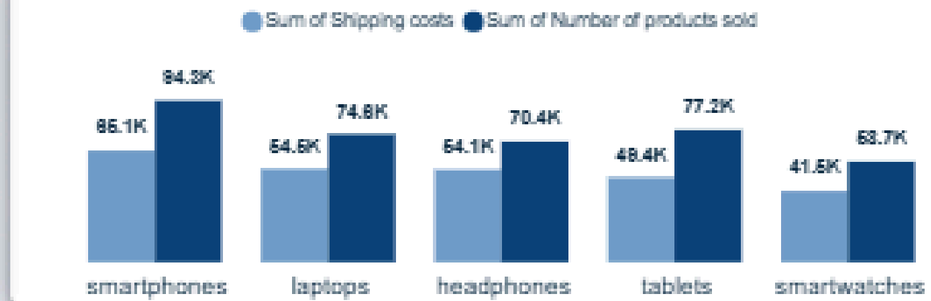
Profit by Delivery Status



On Time Orders and Delayed Orders by Region



Units & Total Shipping Cost by Product Type



SHIPPING & DELIVERY PERFORMANCE

01

1. THE PROFIT GAP

- On-time deliveries (80.4%) generated $71.3M in profit, while delayed orders (19.6%) dropped profits drastically to just $14.3M.

LATE SHIPMENTS CAUSE SIGNIFICANT PROFIT LOSS.

02

2. RISING LOGISTICS COSTS

- Total shipping costs increased to $262.4K (+6.7% vs PY), with the average cost per order rising to $176.0.

COST EFFICIENCY IS DECREASING.

03

3. EFFICIENCY BOTTLENECK

- The East region is the primary bottleneck, suffering the longest average shipping time (7.31 days) and the highest number of delayed orders (67 delays).

EAST REGION UNDERPERFORMS.

04

4. VOLUME DRIVERS

- Smartphones are the highest shipping driver (94,306 units costing $64.0K), while the North region successfully managed the highest volume (78,406 units) with the most on-time deliveries.

NORTH REGION LEADS IN VOLUME HANDLING.

05

5. ACTION ITEM

- Optimize logistics in the East region and implement tighter scheduling to minimize delays, directly capturing the massive $57M profit difference lost to late shipments.

OPTIMIZE REGIONAL OPERATIONS.

RECOMMENDATIONS

SHORT RUN RECOMMENDATIONS

(Operational Actions)

Optimize Discounting Strategy:

Restrict high discount offerings (10%-15% band) as empirical data proves they diminish overall net profit margins; shift focus toward low-to-medium discount bracket (0%-10%) which drive optimal profitability.

Monitor Low-Performing SKUs:

Utilize the tracking of the bottom 5 SKUs by units sold to freeze or eliminate slow-moving stock, effectively reducing holding costs.

Optimize Supplier Sourcing:

Increase procurement shares with Supplier 4 due to their leading Value Score (7.039), while gradually reducing reliance on Supplier 2 due to disproportionately high manufacturing costs relative to quality.

Enforce Strict Quality Control to Reduce Loss:

While supplier reliability significantly improved by 21.19%, stricter quality enforcement is required—particularly for Supplier 5—to curb the 2.1% increase in defect rates.

Elevate Customer Satisfaction:

Work on pushing the stagnating CSAT score (3.28) upward by addressing return root causes, as data proves that product defects, not sales channels or customer demographics, are the primary driver of customer dissatisfaction.

Protect Revenue via On-Time Delivery:

Minimize the current 19.6% delivery delay rate, as on-time fulfillment secures fulfillment secures significantly higher profitability ($71.3M) compared to delayed shipments ($14.3M).

LONG RUN RECOMMENDATIONS

(Strategic Initiatives)

Capitalize on High-Margin Channels:

Scale up operations and marketing for the Online channel to leverage its superior commercial profitability of $18.27M.

Sustain Regional Sales Balance:

Maintain the current distribution equilibrium across all five regions, given the highly stable profit margin performance (ranging between 45% and 48%).

Optimize Return Management:

Develop a structured return management process to reduce return losses and recover product value more efficiently through better inspection, refurbishing, and remarketing strategies.

Invest in “Star” Products:

Increase production support for Smartphones and Headphones, as they simultaneously combine peak sales volumes with top-tier profit margins (above 47%).

Upgrade East Region Logistics:

Restructure the delivery network or sign new courier contracts in the East Region to resolve its high delay rates (67 delayed orders) and longest shipping times (7.31 days).

ONE OF THE KEY RECOMMENDATIONS IS TO INTEGRATE A MACHINE LEARNING MODEL INTO THE SUPPLY CHAIN SYSTEM. THIS REPRESENTS THE NEXT STEP IN EVOLVING THE SOLUTION FROM DESCRIPTIVE ANALYTICS TO INTELLIGENT PREDICTIVE CAPABILITIES. THE DEVELOPED MODEL WILL BE DEMONSTRATED DURING THE PROJECT PRESENTATION.

SPECIAL THANKS TO

ENG. AHMED SAMIR

“

Thank you for all your hard work, support,
and encouragement throughout this journey.
We truly appreciate everything you’ve done for us.
We learned so much from you.

Wishing you continued success!

SPECIAL THANKS TO:

- DEPI

- I LEARN

- EGYPTIAN MINISTRY OF COMMUNICATIONS

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For Your Attention